



What is WiMAX?

WiMAX (Worldwide Interoperability for Microwave access) is touted as the next generation of WiFi, and will be able to connect computers in up to a ten mile range without wires. It can provide connectivity at broadband speeds.

How does WiMAX work?

Imagine a beam of bandwidth that supports data-transmission rates of up to 70 megabits per second within a 30-mile radius of a WiMax antenna—that is basically what WiMAX is. A network provider would set up base stations consisting of one or more sectors that are connected to their edge, and core networks via wireless or wireline connections. In the future, WiMax vendors hope that the technology will be able to communicate with a special chip in wireless devices.

Who is on the WiMAX forum?

The Forum was founded by Nokia, Ensemble Communications and the OFDM Forum, and the membership list includes Aperto Networks, Proxim, Wi-LAN, Intel and Fujitsu.

Why do we need WiMAX?

WiMAX can enable broadband access over entire towns and, with sufficient base stations, entire districts. One good use for WiMAX is to solve the "last mile" problem with cable and DSL—in which transmission from the ISP to the home or office is slowed down due to the wiring.

When will WiMAX reach portable devices?

WiMAX is expected to be incorporated into laptops and PDAs in 2006. Entire cities could become "Metro Zones" for portable outdoor broadband.

Where can I find an instance of WiMAX?

Intel was trying to make Houston County, Georgia, fully connected via WiMax. As of June 2004, the trials were a huge success.

Back To School

Touted as the successor of the extremely popular SDRAM, "DDR"-SDRAM was merely an idea some years ago. Intel had pitched their Rambus RDRAM as the future performance memory type and it seemed like DDR-SDRAM would remain an idea, sans big daddy Intel's backing.

Now, the super fast RDRAM (found in high-end Pentium 4 motherboards) is expensive, and has fallen out of vendors' favour. Double Data Rate Synchronous Dynamic RAM, which costs less, is currently everyone's favourite.

DDR SDRAM helps achieve greater bandwidth than ordinary SDRAM by transferring data on both the rising and falling edges of the clock signal. This doubles the transfer rate without

increasing the frequency of the front side bus. A 100 MHz DDR system, therefore, has an effective clock rate of 200MHz.



DDR RAM provides a memory bus clock rate of *2 (for dual rate) *8 (number of bytes transferred). This basically translates to a transfer rate of 1600 MB/s for a bus frequency of 100MHz.

DDR RAM speed, thus, has a dual nomenclature—that of its effective clock rate (e.g. DDR 200) and its transfer rate (e.g.

though, will obviously vary. For example, Harvard Publications director Peter Kosewski said the university is in a pilot program with Google to scan only 40,000 randomly selected books from its collection of 15 million. By going through the process, Harvard will be able to vet issues such as care of the books and copyright concerns and determine whether it is appropriate to proceed.

Google has long said it plans to make the world's information accessible and searchable, and a cornerstone to its mission would be to bring libraries to life online. Google itself was born out of a library digitisation project at Stanford, Wojcicki said, and its founders had planned all along to build a vast searchable index of books. Only now has the company found the technology and resources to work with libraries to scan their volumes, she said.

Still, the company must navigate tricky issues of copyright. Because libraries own only copies of copyrighted books and don't hold the rights to reproduce those works for wide distribution, Google will likely have to deal with publishers to share revenue on adver-

tising, except only a small portion of material or promote the purchase of books on third-party sites such as Amazon.com, all of which Google said it plans to do. The company said that initially, it will only display biographical information for copyrighted works.

For books in the public domain—books no longer protected by copyright—Google will allow people to search and read the entirety of the work. Oxford, for example, has agreed to let Google scan all its books published in and before 1900.

Many universities tout exclusive collections of books or letters, and for this reason, Google may also run into trouble obtaining clearances down the road to meet its goals. Harvard's Kosewski said that its test is only with a small number of books and that it would require an entirely new set of considerations if the university were to grant Google or others the ability to scan such works.

Rivals are already jockeying for similar utility. Microsoft, for example, has built encyclopaedia answers from its Encarta software into search results for its new proprietary engine. Last year, Yahoo! began a

PC 1600). The suffixed number (DDR 200, DDR 400) designates the speed that the chip is guaranteed to run at. DDR SDRAM DIMMs have 184 pins (SDRAM has 168), and operates at a voltage of 2.5 V (3.3 V for SDRAM)—a factor that can significantly reduce power consumption.

DDR-II has some modifications to allow for higher clock frequencies, but operates on the same principle as DDR. It is now hitting the market. To counter DDR-II, there's Rambus XDR-DRAM. DDR-II may soon become the standard, though, as XDR lacks support. DDR-II is available at clock rates of 400 MHz and higher. AMD has recently introduced support for DDR—the AMD Athlon XP 3200+ requires DDR 400.

content-acquisition project to digitise more searchable material. And Amazon features a search-inside-the-book tool so that people can browse works digitally before buying.

BROWSER WARS II

Firefox On The Hunt

Just as one browser war is barely coming to a boil, the Mozilla Foundation is all set to launch another attack. The fact that Firefox has been downloaded nearly 10 million times since it hit 1.0 seems to have given them the confidence to announce that it will also enter the mobile browser market and take on current leaders Opera.

Why, when Firefox is just picking up steam in the main arena, is there a development team working hard to find another Goliath to battle with? According to Doug Turner, project leader of Minimo, the reasons are clear: Mozilla's browser car-

